Deepak Mallampati

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Kent, OH | US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building healthcare-focused RAG systems, agentic LLM workflows, retrieval/evaluation pipelines, and production inference services. Designed and shipped clinical RAG that lifted retrieval precision ~35% and cut hallucinations >30%; multi-agent healthcare claims pipeline with schema-validated JSON contracts and audit-ready traces; FastAPI/Docker packaging for HIPAA-conscious deployment. Comfortable owning AI features end-to-end - data ingestion, retrieval, agent orchestration, evals, and product integration - in fast-moving generalist roles.

CORE COMPETENCIES

Applied AI • • RAG • • Agentic Workflows • • LLM Tooling • • Retrieval & Evals • • Data Pipelines • • Fine-Tuning & LoRA • • Healthcare AI • • Voice/Multimodal • • FastAPI • • Docker • • Python

PROFESSIONAL EXPERIENCE

Jr. AI/ML Engineer Trainee
Progress Solutions Inc. · USA · Jul 2025 - Present
Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved retrieval precision **~35%** and cut irrelevant retrieval **>30%**; retrieval-grounded response alignment exceeded **90%**.
Developed **agentic LLM workflows** (eligibility checks, care-navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved response stability **~25%**.
Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall **15-20%** on high-risk categories while holding precision **>90%**.
Packaged ML/LLM inference as **FastAPI/Flask** REST services, containerized with **Docker**, structured logs + load simulation; reduced post-deployment defects **~30%**.
Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support tickets; raised dataset reliability above **98%** and cut downstream model instability **~40%**.
Maintained HIPAA-conscious data governance: de-identification, lineage documentation, evaluation audit trails, stakeholder-facing system-limitation docs.
Database & DevOps Performance Engineer (Intern)
Energy Solutions International (Emerson) · Hyderabad, India · Jun 2022 - Apr 2023
Optimized SQL and **T-SQL stored procedures** for the Synthesis Order-to-Cash platform; cut query execution time **~20%** and data retrieval latency **~25%**.
Built **CI/CD pipelines with Jenkins** for schema and stored-procedure deployments; reduced deployment errors **>30%** and release cycle time **~35-40%**.
Designed performance dashboards (SQL Server DMVs, Grafana) for query/deadlock/CPU-I/O monitoring; cut incident recurrence **~25%** and improved root-cause speed **~30%**.
Tuned indexing, statistics, and reconciliation queries; improved reconciliation runtime **~15%**; supported RBAC + audit logging for compliance-sensitive financial modules.
Machine Learning Engineer
Suvidha Foundation · Hyderabad, India · Jan 2022 - Mar 2022
Built a **transformer-based video summarization system** across 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction reduced manual review **60-70%**.
Achieved **~85% alignment** with human-curated highlights; implemented document Q&A with RAG (semantic chunking + embeddings); cut hallucinated outputs **~30%** and raised contextual accuracy **>85%**.
Deployed the stack as a **Flask-based API** with a lightweight web UI for non-technical staff.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with *schema-validated JSON contracts between agents* to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens — Chrome Extension for Real-Time AI Manga/Webtoon Translation *(shipped, Chrome Web Store)*

Full-stack browser extension built independently (Manifest V3, content scripts, service workers, captureVisibleTab). Integrates four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, narrowed host permissions.

Dream Decoder — AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar, Replicate image models). Introduced *intermediate structured prompt transformation layers* — improved contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

Agentic Pixel Character Synthesis & Animation Engine *(ongoing)*

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an *LLM-based agent orchestrator* that decomposes high-level prompts into generation tasks. PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Driver Drowsiness Detection — Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified YOLOv8 detection-and-classification model; reduced inference latency ~30%. LabelImg annotations, augmentation for lighting/head-pose robustness, sliding-window confidence aggregation suppressed blink-driven false positives (~25% reduction), adaptive frame skipping, and NMS tuning for stable real-time operation via OpenCV.

Video Summarization & Highlight Generation System (Suvidha Foundation)

Transformer-based hierarchical abstractive summarization mapped back to source timestamps for automated clip extraction. *60-70% reduction in manual review time*, ~85% highlight alignment with human curation, Flask-based API for non-technical users.

IoT + ML Smart Building Indoor Temperature Prediction System

Arduino + DHT11 + DS3231 sensor pipeline with PLX-DAQ logging; time-series forecasting with walk-forward validation comparing Random Forest, SVR, and Linear Regression. Lag-window feature engineering (n_in=6); Random Forest achieved ~20-30% lower MAE vs linear baseline.

EDUCATION

Master's Degree — Kent State University
Ohio, USA · Aug 2023 - May 2025

CERTIFICATIONS

Open to specialization-specific certifications on request.

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP
LLM & GenAI: LLM apps, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search, responsible AI
ML & Multimodal: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers/Diffusers, LangChain, LlamaIndex, YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics
Data & Infra: Pandas, NumPy, Docker, Jenkins, Grafana, CI/CD, REST APIs, cloud-ready deployment, observability/logging
Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems